

Отчёт о лабораторной работе

Лабораторная работа 2

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1 Цель работы

Приобретение практических навыков по установке и конфигурированию DNS-сервера, усвоение принципов работы системы доменных имён.

2 Выполнение лабораторной работы

Для начала запустим виртуальную машину через vagrant (рис. 2.1).

```
C:\work\dvkalashnikova\code\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
```

Рис. 2.1: Запуск VM

Теперь скачаем пакет bind utils (рис. 2.2).

```

=====
Package                Architecture      Version              Repository           Size
=====
Installing:
bind                   x86_64           32:9.18.33-15.el10_2.10  appstream            323 k
Installing weak dependencies:
bind-dnssec-utils      x86_64           32:9.18.33-15.el10_2.10  appstream            147 k

Transaction Summary
=====
Install 2 Packages

Total download size: 470 k
Installed size: 1.3 M
Downloading Packages:
(1/2): bind-9.18.33-15.el10_2.10.x86_64.rpm          1.3 MB/s | 323 kB    00:00
(2/2): bind-dnssec-utils-9.18.33-15.el10_2.10.x86_64.rpm 552 kB/s | 147 kB    00:00
-----
Total                                                    511 kB/s | 470 kB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Installing     : bind-dnssec-utils-32:9.18.33-15.el10_2.10.x86_64 1/2
  Running scriptlet: bind-32:9.18.33-15.el10_2.10.x86_64           2/2
  Installing     : bind-32:9.18.33-15.el10_2.10.x86_64           2/2
  Running scriptlet: bind-32:9.18.33-15.el10_2.10.x86_64           2/2

Installed:
  bind-32:9.18.33-15.el10_2.10.x86_64          bind-dnssec-utils-32:9.18.33-15.el10_2.10.x86_64

Complete!

```

Рис. 2.2: Скачивание пакетов

Используем команду `dig` для проверки сервисов яндекса (рис. 2.3).

```

complete:
[root@server.dvkalashnikova.net ~]# dig www.yandex.ru

; <<>> DiG 9.18.33 <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43034
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:: udp: 4096
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                187     IN      A      77.88.55.88
www.yandex.ru.                187     IN      A      77.88.44.55
www.yandex.ru.                187     IN      A      5.255.255.77

;; Query time: 29 msec
;; SERVER: 77.88.8.8#53(77.88.8.8) (UDP)
;; WHEN: Tue Sep 15 13:24:34 UTC 2026
;; MSG SIZE rcvd: 90

```

Рис. 2.3: dig ya.ru

Посмотрим на содержание файлов конфигурации dns в etc (рис. 2.4).

```

named.conf /etc/named.conf
[root@server.dvkalashnikova.net ~]# cat /etc/named.conf
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; };

    /*
     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
     - If you are building a RECURSIVE (caching) DNS server, you need to enable
       recursion.
     - If your recursive DNS server has a public IP address, you MUST enable access
       control to limit queries to your legitimate users. Failing to do so will
       cause your server to become part of large scale DNS amplification
       attacks. Implementing BCP38 within your network would greatly
       reduce such attack surface
    */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    geoip-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

```

Рис. 2.4: Файлы конфигурации

Просмотрим теперь файл named.ca (рис. 2.5).

```

[root@server.dvkalashnikova.net ~]# cat /var/named/named.ca
;
; This file holds the information on root name servers needed to
; initialize cache of Internet domain name servers
; (e.g. reference this file in the "cache . <file>"
; configuration file of BIND domain name servers).
;
;
; This file is made available by InterNIC
; under anonymous FTP as
;   file           /domain/named.cache
;   on server      FTP.INTERNIC.NET
; -OR-            RS.INTERNIC.NET
;
; last update:      December 20, 2023
; related version of root zone:  2023122001
;
; FORMERLY NS.INTERNIC.NET
;
;
;      3600000      NS      A.ROOT-SERVERS.NET.
A.ROOT-SERVERS.NET. 3600000  A      198.41.0.4
A.ROOT-SERVERS.NET. 3600000  AAAA    2001:503:ba3e::2:30
;
; FORMERLY NS1.ISI.EDU
;
;      3600000      NS      B.ROOT-SERVERS.NET.
B.ROOT-SERVERS.NET. 3600000  A      170.247.170.2
B.ROOT-SERVERS.NET. 3600000  AAAA    2801:1b8:10::b
;
; FORMERLY C.PSI.NET
;
;      3600000      NS      C.ROOT-SERVERS.NET.
C.ROOT-SERVERS.NET. 3600000  A      192.33.4.12
C.ROOT-SERVERS.NET. 3600000  AAAA    2001:500::c
;
; FORMERLY TERP.UMD.EDU
;
;      3600000      NS      D.ROOT-SERVERS.NET.
D.ROOT-SERVERS.NET. 3600000  A      199.7.91.13
D.ROOT-SERVERS.NET. 3600000  AAAA    2001:500:2d::d
;
; FORMERLY NS.NASA.GOV
;
;      3600000      NS      E.ROOT-SERVERS.NET.
E.ROOT-SERVERS.NET. 3600000  A      192.203.230.10
E.ROOT-SERVERS.NET. 3600000  AAAA    2001:500:a8::e
;
; FORMERLY NS.ISC.ORG
;
;      3600000      NS      F.ROOT-SERVERS.NET.
F.ROOT-SERVERS.NET. 3600000  A      192.5.5.241
F.ROOT-SERVERS.NET. 3600000  AAAA    2001:500:2f::f
;
; FORMERLY NS.NIC.DDN.MIL

```

Рис. 2.5: named.ca

Содержимое named.localhost и named.loopback (рис. 2.6).

```

; End of file[root@server.dvkalashnikova.net ~]# sudo cat /var/named/named.localhost
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
[root@server.dvkalashnikova.net ~]# sudo cat /var/named/named.loopback
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D      ; refresh
                                1H      ; retry
                                1W      ; expire
                                3H )    ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
PTR     localhost.
[root@server.dvkalashnikova.net ~]#

```

Рис. 2.6: named.localhost и named.loopback

Запустим теперь named и осуществим снова dig yandex.ru (рис. 2.7).

```

FIR LOCALHOST.
[root@server.dvkalashnikova.net ~]# systemctl start named
[root@server.dvkalashnikova.net ~]# systemctl enable named
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' → '/usr/lib/systemd/system/named.service'.
[root@server.dvkalashnikova.net ~]# dig www.yandex.ru

; <<> DiG 9.18.33 <<> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39240
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                388     IN      A      77.88.55.88
www.yandex.ru.                388     IN      A      5.255.255.77
www.yandex.ru.                388     IN      A      77.88.44.55

;; Query time: 27 msec
;; SERVER: 77.88.8.8#53(77.88.8.8) (UDP)
;; WHEN: Tue Sep 15 13:29:51 UTC 2026
;; MSG SIZE rcvd: 90

[root@server.dvkalashnikova.net ~]# dig @127.0.0.1 www.yandex.ru
;; communications error to 127.0.0.1#53: timed out

; <<> DiG 9.18.33 <<> @127.0.0.1 www.yandex.ru
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 8931
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 3ab7782010ed7245010000006aa9485cef446931bffe5a6f (good)
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; Query time: 778 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Sep 15 13:30:04 UTC 2026
;; MSG SIZE rcvd: 70

[root@server.dvkalashnikova.net ~]#

```

Рис. 2.7: Запуск named

Теперь настроим порт eth0 (рис. 2.8).

```

[root@server.dvkalashnikova.net ~]# nmcli connection edit eth0

==| nmcli interactive connection editor |==

Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-1x, dcb, sriov, ethtool, match, ipv4, ipv6, prefix-delegation, hostname,
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (31ef6182-2c55-44b3-8944-a2af7d10d59a) successfully updated.
nmcli> quit
[root@server.dvkalashnikova.net ~]# nmcli connection edit System\ eth0
Error: Unknown connection 'System eth0'.
[root@server.dvkalashnikova.net ~]# systemctl restart NetworkManager
[root@server.dvkalashnikova.net ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search dvkalashnikova.net
nameserver 127.0.0.1
nameserver fd17:625c:f037:2::3
[root@server.dvkalashnikova.net ~]#

```

Рис. 2.8: eth0

Откроем и отредактируем named.conf (рис. 2.9).

```

GNU nano 8.1
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secreots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 192.168.0.0/16; };

    /*
     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
     - If you are building a RECURSIVE (caching) DNS server, you need to enable
       recursion.
     - If your recursive DNS server has a public IP address, you MUST enable access
       control to limit queries to your legitimate users. Failing to do so will
       cause your server to become part of large scale DNS amplification
       attacks. Implementing BCP38 within your network would greatly
       reduce such attack surface
    */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    geoip-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

```

Рис. 2.9: named.conf

Установим правила фаервола (рис. 2.10).

```
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dns
success
[root@server.dvkalashnikova.net ~]# firewall-cmd --add-service=dns --permanent
success
[root@server.dvkalashnikova.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
avahi-dae 1120          avahi 12u  IPv4      9118      0t0  UDP *:mdns
avahi-dae 1120          avahi 13u  IPv6      9119      0t0  UDP *:mdns
chronyd  1121          chrony 4u   IPv4     10111     0t0  UDP localhost:323
chronyd  1121          chrony 5u   IPv6     10112     0t0  UDP localhost:323
chronyd  1121          chrony 8u   sock      0,9      0t0  41978 protocol: UDP
wsdd     13585         dvkalashnikova 7u   IPv4     42290     0t0  UDP 239.255.255.250:ws-discovery
wsdd     13585         dvkalashnikova 8u   IPv4     42291     0t0  UDP *:39453
wsdd     13585         dvkalashnikova 9u   IPv4     42292     0t0  UDP server.dvkalashnikova.net:ws-discovery
wsdd     13585         dvkalashnikova 10u  IPv4     42293     0t0  UDP *:39453
```

Рис. 2.10: Правила фаервола

Теперь переместим файл с настройкой конфига (рис. 2.11).

```
[root@server.dvkalashnikova.net ~]# cp /etc/named.rfc1912.zones /etc/named/
[root@server.dvkalashnikova.net ~]# cd /etc/named
[root@server.dvkalashnikova.net named]# mv /etc/named/named.rfc1912.zones /etc/named/dv
[root@server.dvkalashnikova.net named]# nano /etc/named.conf
[root@server.dvkalashnikova.net named]# nano /etc/named/user.net
[root@server.dvkalashnikova.net named]# nano /etc/named/dvkalashnikova.net
```

Рис. 2.11: перемещение файла

И отредактируем наш файл под наши параметры (рис. 2.12).

```

GNU nano 8.1
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//

options {
    listen-on port 53 { 127.0.0.1; any; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secrets";
    recursing-file "/var/named/data/named.recursing";
    allow-query { localhost; 192.168.0.0/16; };

    /*
     - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
     - If you are building a RECURSIVE (caching) DNS server, you need to enable
       recursion.
     - If your recursive DNS server has a public IP address, you MUST enable access
       control to limit queries to your legitimate users. Failing to do so will
       cause your server to become part of large scale DNS amplification
       attacks. Implementing BCP38 within your network would greatly
       reduce such attack surface
    */
    recursion yes;

    dnssec-validation yes;

    managed-keys-directory "/var/named/dynamic";
    geoip-directory "/usr/share/GeoIP";

    pid-file "/run/named/named.pid";
    session-keyfile "/run/named/session.key";

    /* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
    include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/dvkalashnikova.net";

```

Рис. 2.12: Редактирование файла

То же самое сделаем с файлом зон (рис. 2.13).

```
GNU nano 8.1
// named.rfc1912.zones:
//
// Provided by Red Hat caching-nameserver package
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc6303
// (c)2007 R W Franks
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//
zone "dvkalashnikova.net" IN {
    type master;
    file "master/fz/dvkalashnikova.net";
    allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
    type master;
    file "master/rz/192.168.1";
    allow-update { none; };
};
```

Рис. 2.13: Файл зон

Создадим папки с настройками днс (рис. 2.14).

```
[root@server.dvkalashnikova.net named]# cd /var/named
[root@server.dvkalashnikova.net named]# mkdir -p /var/named/master/fz
[root@server.dvkalashnikova.net named]# mkdir -p /var/named/master/rz
[root@server.dvkalashnikova.net named]# cp /var/named/named.localhost /var/named/master/fz/
[root@server.dvkalashnikova.net named]# cd /var/named/master/fz/
[root@server.dvkalashnikova.net fz]# mv named.localhost dvkalashnikova.net
[root@server.dvkalashnikova.net fz]# nano /var/named/master/fz/dvkalashnikova.net
```

Рис. 2.14: Создание папок и настроек днс

Отредактируем файл dvkalashnikova.net (рис. 2.15).

```

GNU nano 8.1
$TTL 1D
@      IN SOA  @ server.dvkalashnikova.net. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      192.168.1.1
AAAA   ::1
$ORIGIN dvkalashnikova.net.
server A 192.168.1.1
ns     A 192.168.1.1

```

Рис. 2.15: dvkalashnikova.net

Теперь посмотрим на файлы из папки rz (рис. 2.16).

```

[root@server.dvkalashnikova.net fz]# cp /var/named/named.loopback /var/named/master/rz/
[root@server.dvkalashnikova.net fz]# cd /var/named/master/rz/
[root@server.dvkalashnikova.net rz]# mv named.loopback 192.168.1
[root@server.dvkalashnikova.net rz]# nano

```

Рис. 2.16: Папка rz

Отредактируем следующим образом (рис. 2.17).

```

GNU nano 8.1
$TTL 1D
@      IN SOA  @ rname.invalid. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum

NS     @
A      127.0.0.1
AAAA   ::1
PTR    localhost.

```

Рис. 2.17: Редактирование файла

Настроим Selinux (рис. 2.18).

```
[root@server.dvkalashnikova.net rz]# chown -R named:named /etc/named
[root@server.dvkalashnikova.net rz]# chown -R named:named /var/named
[root@server.dvkalashnikova.net rz]# restorecon -vR /etc
[root@server.dvkalashnikova.net rz]# restorecon -vR /var/named
[root@server.dvkalashnikova.net rz]# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
[root@server.dvkalashnikova.net rz]# setsebool named_write_master_zones 1
[root@server.dvkalashnikova.net rz]# setsebool -P named_write_master_zones 1
[root@server.dvkalashnikova.net rz]#
```

Рис. 2.18: Selinux

Через dig попробуем подключиться к собственному днс (рис. 2.19).

```
[root@server.dvkalashnikova.net rz]# dig ns.dvkalashnikova.net

; <<>> DiG 9.18.33 <<>> ns.dvkalashnikova.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 14760
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 6bab7019444d8057010000006aa9533be773cfdb604e7b12 (good)
;; QUESTION SECTION:
;ns.dvkalashnikova.net.      IN      A

;; ANSWER SECTION:
ns.dvkalashnikova.net.  86400  IN      A      192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Tue Sep 15 14:16:27 UTC 2026
;; MSG SIZE rcvd: 94

[root@server.dvkalashnikova.net rz]# host -l dvkalashnikova.net
dvkalashnikova.net name server dvkalashnikova.net.
dvkalashnikova.net has address 192.168.1.1
dvkalashnikova.net has IPv6 address ::1
ns.dvkalashnikova.net has address 192.168.1.1
server.dvkalashnikova.net has address 192.168.1.1
[root@server.dvkalashnikova.net rz]# host -a dvkalashnikova.net
Trying "dvkalashnikova.net"
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 59560
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;dvkalashnikova.net.      IN      ANY

;; ANSWER SECTION:
dvkalashnikova.net.  86400  IN      SOA     dvkalashnikova.net. server.dvkalashnikova.net. 0 86400 3600 604800 10800
dvkalashnikova.net.  86400  IN      NS      dvkalashnikova.net.
dvkalashnikova.net.  86400  IN      A       192.168.1.1
dvkalashnikova.net.  86400  IN      AAAA    ::1

Received 137 bytes from 127.0.0.1#53 in 0 ms
[root@server.dvkalashnikova.net rz]# host -t A dvkalashnikova.net
dvkalashnikova.net has address 192.168.1.1
[root@server.dvkalashnikova.net rz]# host -t PTR 192.168.1.1
1.1.168.192.in-addr.arpa domain name pointer ns.dvkalashnikova.net.
1.1.168.192.in-addr.arpa domain name pointer server.dvkalashnikova.net.
[root@server.dvkalashnikova.net rz]#
```

Рис. 2.19: dig

Оформим нашу работу как конфигурацию для вагранта (рис. 2.20).

```
[root@server.dvkalashnikova.net rz]# cd /vagrant
[root@server.dvkalashnikova.net vagrant]# mkdir -p /vagrant/provision/server/dns/etc/named
[root@server.dvkalashnikova.net vagrant]# mkdir -p /vagrant/provision/server/dns/var/named/master/
[root@server.dvkalashnikova.net vagrant]# cp -R /etc/named.conf /vagrant/provision/server/dns/etc/
[root@server.dvkalashnikova.net vagrant]# cp -R /etc/named/* /vagrant/provision/server/dns/etc/named/
[root@server.dvkalashnikova.net vagrant]# cp -R /var/named/master/* /vagrant/provision/server/dns/var/named/mast
[root@server.dvkalashnikova.net vagrant]# touch dns.sh
[root@server.dvkalashnikova.net vagrant]# chmod +x dns.sh
[root@server.dvkalashnikova.net vagrant]# nano dns.sh
```

Рис. 2.20: Конфиг вагрант

И напишем скрипт для загрузки вагранта (рис. 2.21).

```
GNU nano 8.1
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install bind bind-utils
echo "Copy configuration files"
cp -R /vagrant/provision/server/dns/etc/* /etc
cp -R /vagrant/provision/server/dns/var/named/* /var/named
chown -R named:named /etc/named
chown -R named:named /var/named
restorecon -vR /etc
restorecon -vR /var/named
echo "Configure firewall"
firewall-cmd --add-service=dns
firewall-cmd --add-service=dns --permanent
echo "Tuning SELinux"
setsebool named_write_master_zones 1
setsebool -P named_write_master_zones 1
echo "Change dns server address"
nmcli connection edit "System eth0" <<EOF
remove ipv4.dns
set ipv4.ignore-auto-dns yes
set ipv4.dns 127.0.0.1
save
quit
EOF
systemctl restart NetworkManager
echo "Start named service"
systemctl enable named
systemctl start named
```

Рис. 2.21: скрипт

И в vagrantfile будем загружать этот скрипт (рис. 2.22).

```
server.vm.provision "server dummy", type: "shell", preserve_order: true, path: "provision/server/01-dummy.sh"
server.vm.provision "server forward", type: "shell", preserve_order: true, run: "always", path: "provision/server/02-forward.sh"
server.vm.provision "server dns", type: "shell", preserve_order: true, path: "provision/server/dns.sh"
end
```

Рис. 2.22: vagrantfile

3 Выводы

В результате выполнения работы были получены навыки настройки днс